



Task 1 — Exploratory Data Analysis

Uber Fare Prediction Project • ML Internship Program

■ Objective

Before building any model, you need to deeply understand your data. This task trains you to explore, question, and visualize data — and to decide **which plot best answers each question**. The insights you find here will directly shape your decisions in Task 2 (modeling).

■ Dataset Description

The dataset contains real-world Uber ride records from New York City. Each row represents one trip. Below is a description of every column — read this carefully before starting your analysis.

Dataset Link :

[final_internship_data.csv - Google Drive](#)

Column	Type	Description
User_ID	Integer	Unique identifier per user — not useful for modeling
User_Name	String	Passenger name — not useful for modeling
Driver_Name	String	Driver name — not useful for modeling
Car_Condition	Categorical	Vehicle condition: Good / Bad
Weather	Categorical	Weather at pickup: Sunny / Windy / Winter
Traffic_Conditions	Categorical	Traffic level: Normal / Heavy
Key	Integer	Unique trip ID — use only for deduplication
Fare_Amount	Float	■ TARGET — the fare paid in USD
Pickup_Datetime	DateTime	Timestamp when the ride was requested
Pickup_Longitude	Float	Longitude of pickup location
Pickup_Latitude	Float	Latitude of pickup location
Dropoff_Longitude	Float	Longitude of drop-off location

Dropoff_Latitude	Float	Latitude of drop-off location
Passenger_Count	Integer	Number of passengers in the ride
Hour	Integer	Hour extracted from Pickup_Datetime (0–23)
Day	Integer	Day of week (0=Monday ... 6=Sunday)
Month	Integer	Month of pickup (1–12)
Week	Integer	Week number of the year
JFK_Dist	Float	Distance from pickup to JFK Airport (km)
EWB_Dist	Float	Distance from pickup to EWR Airport (km)
LGA_Dist	Float	Distance from pickup to LGA Airport (km)
SOL_Dist	Float	Distance from pickup to Statue of Liberty (km)
Distance	Float	Total trip distance (km)
Bearing	Float	Trip direction angle in degrees (0–360)

■ Which Plot Should You Use? — Quick Reference

Choosing the right visualization is a core data science skill. This table summarizes the most common EDA plots and when to use each one. Refer to it when answering the training questions below.

Plot	Best For	Example Question	Limitation
Histogram	Distribution of 1 numeric variable	How is Fare_Amount distributed?	Sensitive to bin size
Boxplot	Distribution + outliers of numeric variable	Are there extreme fares?	Hides sample size
Bar Chart	Compare averages across categories	What is the avg fare per Weather type?	Hides variance within groups
Line Chart	Trends over ordered values (time)	How does avg fare change by hour?	Only for ordered/time data
Scatter Plot	Relationship between 2 numeric variables	Does Distance predict Fare?	Slow on large datasets
Heatmap (corr.)	Correlations between all numeric features	Which features relate most to Fare?	Only linear relationships
Heatmap (pivot)	Two-variable patterns (e.g., time x day)	When is demand highest?	Needs aggregation first
Count Plot	Frequency of categories	How many rides per Traffic type?	Doesn't show fare
KDE Plot	Smooth distribution — compare groups	Does fare differ by Car_Condition?	Harder to read exact values
Geo Scatter	Spatial distribution of pickup points	Where in NYC are rides concentrated?	Needs valid coordinates
Pair Plot	All pairwise relationships at once	Quick overview of feature relationships	Slow on many columns

■ Training Questions — Choose the Right Plot

For each question below, first identify **which plot is most appropriate and why**, then write the code to produce it. There may be more than one valid answer — justify your choice.

Q1) Is higher Passenger higher price?

Passenger count : Numerical

Price : numerical

Suitable plot : scatterplot

What you will see : whether there are relationship by closeness of points or not

Q2) Does car condition influence the fare amount?

Car condition : categorical

Fare : numerical

Suitable plot : boxplot/ barplot

Q3) At what hour of the day are fares typically highest?

Hour : numerical

Fare : numerical

Suitable plot : lineplot

Q4) Does traffic condition affect the trip fare?

- **Traffic_Conditions:** Categorical
- **Fare_Amount:** Numerical
- **Suitable plot:** Boxplot / Barplot
- **What you will see:** Whether rides during heavy traffic tend to have higher fares compared to normal traffic.

Q5) Is trip distance the strongest predictor of the fare amount?

- **Distance:** Numerical
- **Fare_Amount:** Numerical
- **Suitable plot:** Scatter Plot
- **What you will see:** Whether fare increases as trip distance increases and whether the relationship appears linear.

Q6) At what time of day are ride requests most frequent?

- **Hour:** Numerical (Discrete)
- **Ride Count:** Frequency
- **Suitable plot:** Count Plot / Bar Chart
- **What you will see:** Peak demand hours and periods with the highest number of trips.

(Different from your current question "highest fare by hour"; this one analyzes demand rather than price.)

Q7) Does weather condition influence the average trip distance?

- **Weather:** Categorical
- **Distance:** Numerical
- **Suitable plot:** Boxplot / Barplot
- **What you will see:** Whether people tend to travel longer or shorter distances under different weather conditions.

Q8

Are rides that start closer to airports generally more expensive?

- **JFK_Dist (or EWR_Dist / LGA_Dist):** Numerical
- **Fare_Amount:** Numerical
- **Suitable plot:** Scatter Plot
- **What you will see:** Whether proximity to airports is associated with higher fares.

■ Extra Resources

	Resource
1	https://youtu.be/oOv1gE0ER3I?si=pF7bTLFAMhedMJxF
2	https://youtu.be/oCXfRe86dbA?si=EgfKP2c6FuWBK_HE
3	https://youtu.be/VaSjiJMrq24?si=gyf7YrCaA0kSNjl
4	https://youtu.be/rGx1QNdYzs?si=k94xupWSmrXooWn1
	Websites To Learn Visualization
1	https://datavizcatalogue.com/
2	https://datavizproject.com/#

■ Submission Checklist

- Benefit from above questions to analyze and create **new Questions for understand the data**
- Notebook contain : question, plot choice, cleaning notes, code, and interpretation
- Presentation shows : question, plot and the insights in story
- Code is clean and commented

Remember: The goal is not to produce the most code — it is to produce the most insight. A single

